

Standardization documents — Document metamodel

Working Draft Standard

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Contents

Foreword.....	v
0. Introduction.....	vi
1. Scope.....	1
2. Normative References.....	1
3. Terms and definitions.....	1
4. Architecture.....	3
4.1. General.....	3
4.2. Dependent data models.....	3
4.3. Structure.....	3
4.4. Underlying document model.....	3
4.5. Bibliographic and citation models.....	3
4.6. Hierarchical containment.....	4
4.7. Document model specialization.....	4
5. StandardDocument.....	5
5.1. BasicDocument.....	5
5.2. StandardDocument.....	5
5.3. BoilerplateType.....	6
6. Document metadata elements.....	6
6.1. Purpose.....	6
6.2. Structure.....	7
6.3. BibDataExtensionType.....	9
6.4. BibData.....	9
6.5. StandardBibData.....	9
6.6. StandardBibDataExtensionType.....	9
7. Structured identifier.....	9
7.1. StructuredIdentifierType.....	10
8. Section elements.....	11
8.1. Purpose.....	11
8.2. Structure.....	11
8.3. Clause sections.....	11
8.4. Content sections.....	11
8.5. Annex sections.....	11
8.6. Category label.....	12
8.7. Restrictions on hanging elements.....	12
8.8. Overview.....	13
9. Citation and reference elements.....	14
9.1. Purpose.....	14
9.2. References section.....	14
9.3. StandardReferencesSection.....	14
9.4. BibliographicItem.....	14
9.5. Citation.....	14
9.6. Reference item.....	14
9.7. Citation.....	15
10. Terms and definition elements.....	15
10.1. Purpose.....	15
10.2. General.....	15
10.3. Definition sections.....	15
10.4. StandardReferencesSection.....	15
10.5. BibliographicItem.....	16
10.6. Citation.....	16
10.7. Term sections.....	17
10.8. Term.....	17
10.9. TermCollection.....	17
10.10. Term.....	18
10.11. RelatedTerm.....	19

10.12	RelatedTermType	19
10.13	TermSource	19
10.14	Nested sections	19
10.15	Overview	20
11.	Inline elements	21
12.	Block	21
13.	Data types	21
Annex A. (informative)	Profile: ISO standardization deliverables	22
Annex B. (informative)	Profile: CalConnect standardization deliverables	31
Annex C. (informative)	Concept and concept collection models	35
Bibliography		47

Foreword

The Calendaring and Scheduling Consortium (“CalConnect”) is a global non-profit organization with the aim to facilitate interoperability of collaborative technologies and tools through open standards.

CalConnect works closely with international and regional partners, of which the full list is available on our website (<https://www.calconnect.org/about/liaisons-and-relationships>).

The procedures used to develop this document and those intended for its further maintenance are described in the CalConnect Directives.

In particular the different approval criteria needed for the different types of CalConnect documents should be noted. This document was drafted in accordance with the editorial rules of the CalConnect Directives.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CalConnect shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be provided in the Introduction.

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

This document was prepared by Technical Committee *VCARD*.

0. Introduction

0.1.. General

Standardization documents are documents created to express a set of normalized requirements, often themselves created through a process of standardization.

Standardization documents play a crucial role in today's activities. Rule-based enforcements from the organization of markets, conformance of products to the compliance of regulation and requirements, all depend on content expressed through standardization documents.

0.2.. Challenges

The process of generating standardization documents is increasingly problematic due to the increased work burden on standardization bodies.

Often, concerns are raised over both the efficiency and the effectiveness of the workflows around generating standardization documents in those bodies and elsewhere. Some practices include:

- Rigorous and detailed rules around document structure articulated by standardization bodies are being manually validated by the editorial teams, with mixed and often inconsistent results.
- Editing of documents substantially revolves around standardizing use of particular software, instead of requirements, with all the problems that conflating document structure and presentation introduce.
- The authoring and editorial processes spend significant amounts of time on formatting rather than document content, and is vulnerable to error.
- Existing exchange mechanisms for documents have been claimed to focus on document appearance, rather than document structure, and so do not alleviate the formatting burden on the editorial process.
- Standardization bodies often demand vastly different workflows that take different inputs. Submitting identical content or cross-publishing to more than one standardization body requires replicating a significant amount of work.

0.3.. Data-centric approach

The *StandardDocument* document metamodel aims to treat a standardization document as a set of standardized data structures, used to enable an end-to-end standardization document production process.

The process is centered around the *StandardDocument* document model, designed to focus on semantics and encompasses core similarities between standardization documents of different standardization bodies.

This approach has the following benefits:

- The approach enables an end-to-end, author-to-publishing workflow, with well-managed data artefacts along each stage, that editorial bodies can exercise appropriate control over. This allows document management processes to be centered around data.
- The semantic nature of the document metamodel (and its specialized models) abstracts away presentation, such as layout and formatting. This allows the authoring and editorial to focus on content.
- The structural and content requirements around standardization documents can be fully captured in the document model, and can therefore be subject to formal validation.
- The document model can be highly similar across different standardization documents, and therefore the content in such documents can be easily repurposed for different standardization bodies with minimal effort.

0.4.. Compatibility

The document metamodel in this standard has been designed to reflect the needs and expressiveness of existing standardization processes, requirements and outputs of the following standardization bodies:

- ISO and IEC ([ISO/IEC DIR 1](#), [ISO/IEC DIR 2:2018](#))
- ITU (TTP, ATP)
- CalConnect ([CC/DIR 10001:2019](#), [CC/DIR 10002:2019](#))
- IETF ([RFC 7749](#), [RFC 7991](#))

- OGC ([OGC 05-020r26](#))
- NIST Special Publications
- China standards (national, local, industry, social group) ([GB/T 1.1-2009](#))

In many cases, the document model also enables alignment with capabilities of existing document production tools. The document metamodel has already been adopted by a number of standardization bodies listed above.

Standardization documents — Document metamodel

1. Scope

This document provides a document metamodel for standardization documents and describes its alignment to standardization processes.

The following aspects of the document metamodel are out of scope:

- implementation of the document metamodel into a document model;
- derivation and mapping of the reference model to representation formats (such as for serialization); and
- prescription of profiles of the document metamodel for particular standardization bodies.

Example profiles provided for ISO and CalConnect standards in [Annex A](#) and [Annex B](#) are presented for informative reasons.

2. Normative References

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 639 (all parts), *Code for the representation of names of languages*

ISO 8601-1, *Date and time – Representations for information interchange – Part 1: Basic rules*

ISO/IEC 10118 (all parts), *Information technology – Security techniques*

ISO/IEC 14888 (all parts), *Information technology – Security techniques*

ISO 15924:2004, *Information and documentation – Codes for the representation of names of scripts*

ISO/IEC Guide 2 (all parts), *Standardization and related activities – General vocabulary*

RFC 3986, *Uniform Resource Identifier (URI): Generic Syntax*

3. Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1.

standardizing body

body that has recognized activities in standardization

[SOURCE: ISO/IEC Guide 2 (all parts)]

3.2.

document

medium with information recorded on or in it

[SOURCE: ISO/IEC Guide 2 (all parts)]

3.3.

standardization document

document that provides rules, guidelines or characteristics for activities or their results

Note 1 to entry: Documents such as standards, technical specifications, codes of practice and regulations are considered part of this term.

Note 2 to entry: This terms closely corresponds to “normative document” defined in [ISO/IEC Guide 2 \(all parts\)](#).

3.4.

class

structure containing a description of an entity in terms of its components

3.5.

subclass

class (class) which inherits from another class its component descriptions, and optionally adds to them its own component descriptions

3.6.

document metamodel

formal specification of the structure of a document model in terms of its components and their arrangement, expressed through *class* (classes)

3.7.

document model

formal specification of the structure of a document in terms of its components and their arrangement, expressed through *class* (classes)

3.8.

standards model

document model (document model) for *standardization document* (standardization documents)

3.9.

paragraph

subdivision of running text, normally run on throughout, that is separated from text before and after by a change of line and stands below any chapters or *section* (sections)

[SOURCE: ISO 5127:2017, Clause 3.5.8.07]

3.10.

block

grouping of text at the *paragraph* (paragraph) level

3.11.

inline element

grouping of text that can be contained within a *paragraph* (paragraph), including plain strings

3.12.

section

hierarchical subdivision of a document, consisting of one or more *block* (blocks), and/or one or more sections

3.13.

identifier

a character, or group of characters, used to identify or name an item of data and possibly to indicate certain properties of that item

[SOURCE: IEC 60050:2011]

3.14.

term

expression with a particular meaning that is part of the specialized vocabulary of a field

4. Architecture

4.1. General

The *StandardDocument* document metamodel is a semantically oriented data model that depends on an underlying document model for implementation.

StandardDocument is closely aligned with the *BasicDocument* model ([CC 36010](#)), but it can be implemented on top of any document model that is compatible with the *BasicDocument* model or satisfies the requirements described in this document.

4.2. Dependent data models

The implementation of the document metamodel relies on the availability of the following data models:

- A document data model that supports modeling of textual content at the block and inline levels, as well as modifications to the document itself.
- Data models for representing bibliographic information and citations for a document.

In this document, the *BasicDocument* document model from [CC 36010](#) is used as an exemplary document model, the *BibliographicItem* and its related data models from [CC 6900](#) are used for representing bibliographic and citation models.

4.3. Structure

The *StandardDocument* document metamodel is expressed hierarchically in the following structure:

- Document representation: the underlying document model, described as *BasicDocument* in this document, is used to express the structure of generic documents (with specific requirements detailed in the *BasicDocument* model from [CC 36010](#));
- Standardization document structure: the *StandardDocument* model is specialized from *BasicDocument*, conveying the structure particular to standardization documents as a class.
- Specific types of standardization documents: the *StandardDocument* model is further specialized into models specific to particular standards. Specific standards models can form the basis in turn of other standards models.

4.4. Underlying document model

The underlying document model must provide the following classes of document element data models:

- Sections. A grouping of elements that can be shown under the table of contents (see the *Section* models in [CC 36010](#))
- Blocks. A grouping of paragraphs or paragraph-levels of text (see *Block* models in [CC 36010](#))
- Inline elements. Groupings of text smaller than a paragraph, including plain strings. (see *InlineElement* and related models in [CC 36010](#))

4.5. Bibliographic and citation models

Standardization documents often contain references to and citations of external information resources, as well as bibliography sections.

The *StandardDocument* model depends on data models that can fully represent:

- metadata of referenced information resources;
- presentation of a referenced information resource; and

- locality of a citation.

For example, if a standardization document needs to adhere to the citation guidelines of [ISO 690 \(all parts\)](#), data models that facilitate these rules such as [CC 6900](#) should be used.

The `BibliographicItem`, `Citation` and related models from [CC 6900](#) are used in the reference metamodel for its expression of bibliographic references and citations.

4.6. Hierarchical containment

In the document metamodel, the classes are in a strict hierarchical relation:

- Documents consist of sections, which consist of blocks, which consist of inline elements.
- Sections can be nested within sections (e.g. clauses and subclauses).
- Blocks can be nested within blocks (e.g. nested lists).
- Inline elements can be embedded within other inline elements (e.g. bold with italics).

Sections in the *StandardDocument* model are not expected to be siblings of blocks, nor blocks of inline elements.

EXAMPLE

A list (block) is not expected to occur next to inline text within a paragraph. For that reason, paragraphs cannot contain other block elements, such as lists or tables.

NOTE 1 This constraint is adopted from [CC 36010](#), and is not present in XML-based schemas like HTML, TEI-C, or DocBook. It confers the advantage of having a significantly streamlined document model. While this limits some expressive potential, the difference is minor and semantically desired, particularly with regards to the unambiguous rendering of paragraphs.

4.7. Document model specialization

StandardDocument can be specialized in the following ways:

- specializing classes from the `BasicDocument` models;
- specializing classes from the `StandardDocument` models;
- adding new classes;
- changing attributes of a class.

The specialization method is not restricted to adding attributes, as is the case in typical entity subclassing; it can also include removing attributes from a class, changing their obligation and cardinality, and changing their type, including changing enumerations. Attributes can be overruled at any level; for example, standards-specific models routinely enhance the bibliographic model at the base of the hierarchy.

For reasons of clarity, however, renaming classes and attributes should be avoided in specialization.

EXAMPLE

The document model for the Chinese National Standard (“GB Standard”), as described in [GB/T 1.1-2009](#), is structurally modelled after ISO documents, explicitly described as a localization of [ISO/IEC DIR 2:2004](#). Therefore, the Chinese National Standard model is best expressed as a specialization of the ISO Standards model.

5. StandardDocument

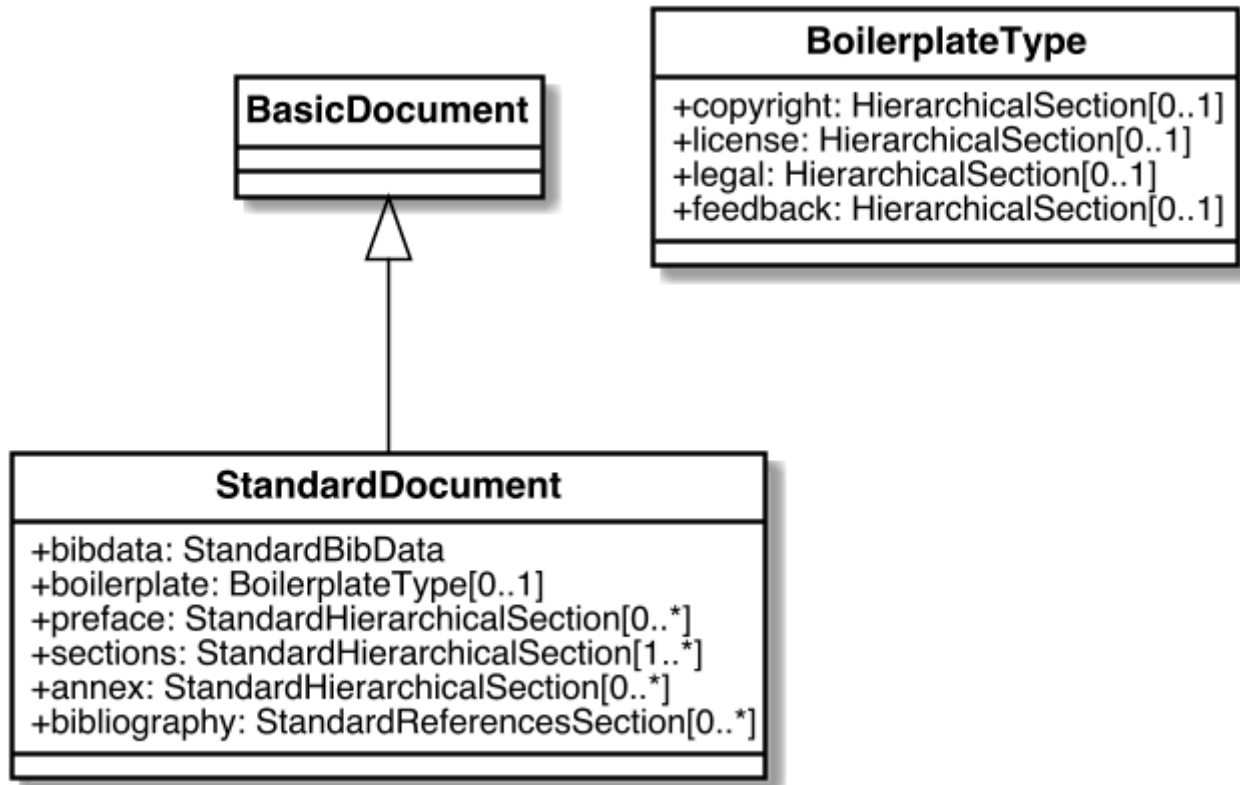


Figure 1 — Standard document class data model

5.1. BasicDocument

5.2. StandardDocument

The StandardDocument model is used to represent a standardization document.

Typically, a standardization document contains at least the following data elements:

- Metadata information;
- Clauses and subclauses;
- Bibliographies;
- Annexes if applicable.

The exact component requirements of a standardization document can differ widely from one standardization body to the next. Specialization is necessary to adopt the StandardDocument model for such standardization bodies.

The StandardDocument model extends the BasicDocument modelling of the document by requiring specific types of sections.

Table 1 — StandardDocument attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
bibdata	TODO: enum bibdata's definition	M	1	StandardBibData

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
boilerplate	An optional <i>boilerplate</i> section, intended to appear at the front of the document. It consists of content addressing <i>copyright</i> , <i>license</i> , <i>legal</i> , and <i>feedback</i> concerns	O	1	BoilerplateType
preface	Zero or more optional <i>preface</i> sections (each as a <code>StandardHierarchicalSection</code> , Clause 8)	O	N	StandardHierarchicalSection
sections	One or more <i>sections</i> (each as a <code>StandardHierarchicalSection</code>)	M	N	StandardHierarchicalSection
annex	Zero or more <i>annexes</i> (each as a <code>StandardHierarchicalSection</code>)	O	N	StandardHierarchicalSection
bibliography	Zero or more <i>bibliographies</i> (each as a <code>StandardReferencesSection</code> , Clause 9)	O	N	StandardReferencesSection

5.3. BoilerplateType

Table 2 — BoilerplateType attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
copyright	TODO: enum copyright's definition	O	1	HierarchicalSection
license	TODO: enum license's definition	O	1	HierarchicalSection
legal	TODO: enum legal's definition	O	1	HierarchicalSection
feedback	TODO: enum feedback's definition	O	1	HierarchicalSection

6. Document metadata elements

6.1. Purpose

Document metadata of a standardization document is used to provide basic information of the standardization document.

This includes data elements such as:

- title(s);
- document identifier(s);
- authorship and contribution information;
- publisher and distributor details;
- rights information;
- document development stage;
- relevant dates; and
- edition.

6.2. Structure

Document metadata in *StandardDocument* is handled using the `StandardBibliographicItem` class extended from the `BibliographicItem` class defined in [CC 6900](#).

In standardization documents, certain data elements, such as document identifiers and document development stage information, are often highly specific to a standardization body.

Therefore, based on `BibliographicItem`, `StandardBibliographicItem` implements the following extension points:

- the document type (already provided in the `BasicDocument` model), it is assumed that particular specializations of the document metamodel will substitute their own enumerations of particular subclasses of document, which will be more granular;
- zero or more structured identifiers ([Clause 7](#)).

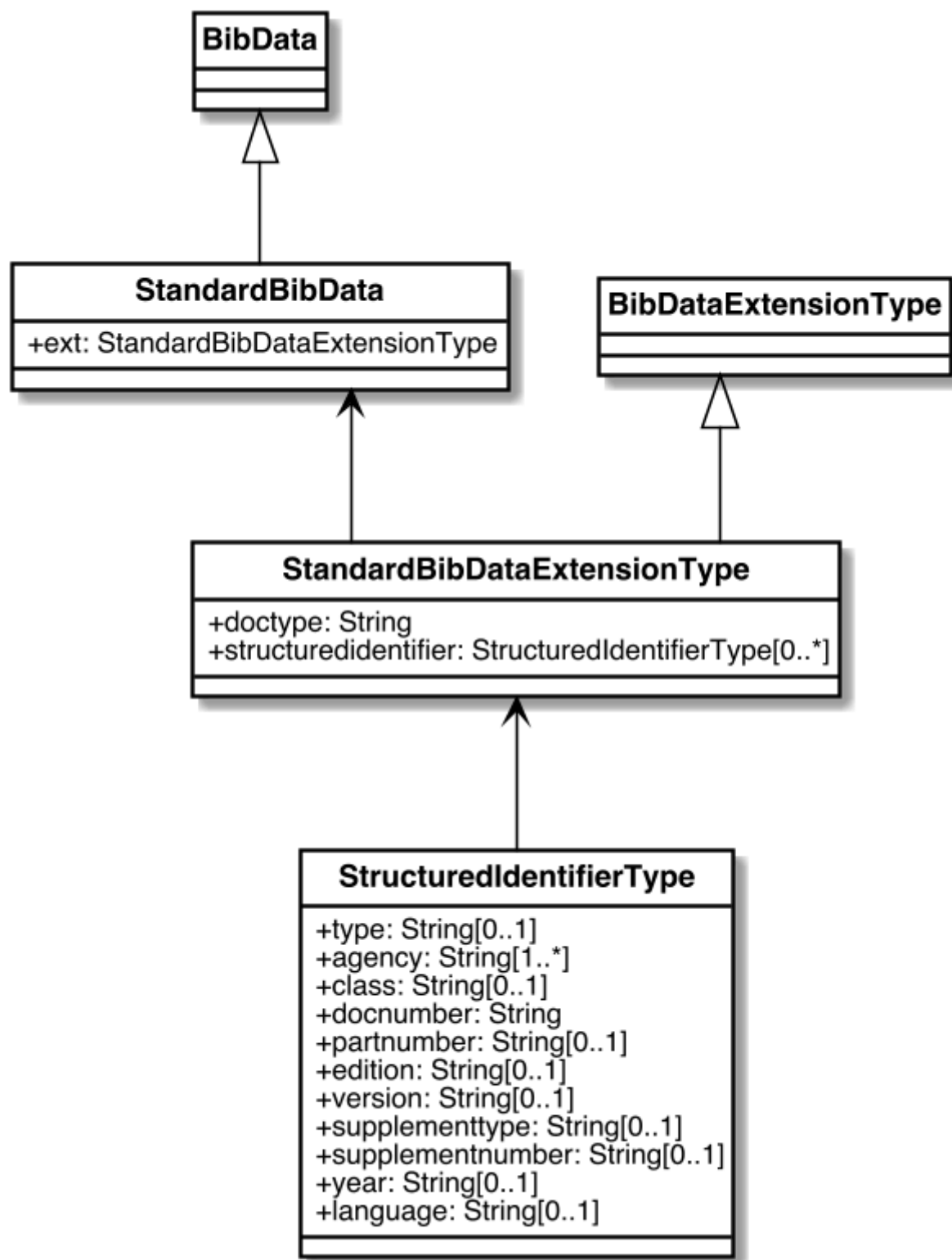


Figure 2 — Metadata class data models

6.3. BibDataExtensionType

6.4. BibData

6.5. StandardBibData

Table 3 — StandardBibData attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
ext	TODO: enum ext's definition	M	1	StandardBibDataExtensionType

6.6. StandardBibDataExtensionType

Table 4 — StandardBibDataExtensionType attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
doctype	TODO: enum doctype's definition	M	1	String
structuredidentifier	TODO: enum structuredidentifier's definition	O	N	StructuredIdentifierType

7. Structured identifier

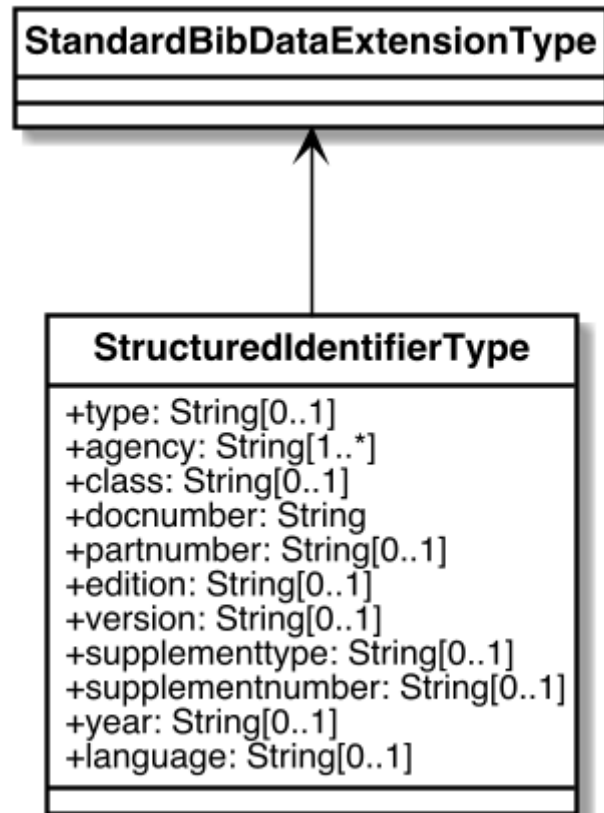


Figure 3 — Structured identifier class data model

7.1. StructuredIdentifierType

Standardization bodies often utilize structured, canonical identifiers according to a schema.

Various components of this identifier is meant to be semantically marked up.

The structure given here is meant to be applicable across standardization bodies, but it can always be customized for a specific standardizing body.

This information should all be represented elsewhere in the document metadata, but is replicated in the identifier.

Table 5 — StructuredIdentifierType attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	type of identifier, used to differentiate different instances of structured identifier generated for different purposes	O	1	String
agency	one or more abbreviations identifying the standardization body involved in producing this standard, and treated as a namespace for the identifier	M	N	String
class	an optional abbreviation identifying the subclass of standard or other document	O	1	String
docnumber	a number or other identifier for the document, issued by the standardizing body	M	1	String
partnumber	an optional part number for the document; includes sub-part numbers if applicable	O	1	String
edition	an optional string giving the edition of the standard	O	1	String
version	an optional string giving the version of the standard (within the edition). Includes the document stage that a draft has reached, e.g. “FDIS” under ISO	O	1	String
supplementtype	an abbreviation for the type of supplement to a standard that this document represents, if applicable	O	1	String
supplementnumber	an abbreviation for the number or other primary identifier of supplement to a standard that this document represents, if applicable	O	1	String
year	an optional string giving the date (typically year) that this particular edition of the standard was published	O	1	String
language	an optional abbreviation identifying the language of the standard	O	1	String

8. Section elements

8.1. Purpose

A section represents a clause in a standardization document. Specifically, a section represents a logically coherent group of paragraphs and blocks that is meant to be shown in the table of contents.

8.2. Structure

The `StandardDocument` model is a specialization of the `BasicSection` class of the `BasicDocument` model by as follows:

- All sections in a `StandardDocument` model are instances of the `StandardSection` class, which is a subclass of `BasicSection`.
- The `StandardSection` class indicates its status as normative or informative explicitly, through the `status` attribute. The normative status of individual sections in a standardization document is pertinent.

EXAMPLE

The necessity of indicating normative or informative status is specified in [ISO/IEC DIR 2:2018](#) for ISO deliverables.

There are two major types of sections, namely, clause sections and content sections.

8.3. Clause sections

Clause sections (`ClauseSection`) are numbered, and form part of the main flow of the document.

Clause sections are by default hierarchical (as an exception, see [Clause 10](#)). The subclauses of Clauses are themselves Clause Sections; they continue the hierarchical numbering of Clause Sections at deeper hierarchical levels.

8.4. Content sections

Content sections (`StandardHierarchicalSection`) are optionally not numbered, and lie outside the main flow of the document.

In `StandardDocument`, there are four distinctly named content sections. These content sections are designated sections as they recur in standardization documents and have distinct semantics:

- Abstract,
- Foreword,
- Introduction, and
- Acknowledgments.

Specializations of the model may define other such sections.

The subclauses of Content sections are `StandardSections`, and need not be numbered.

8.5. Annex sections

Annexes are implemented as `ClauseSections` in the `StandardDocument` metamodel.

They are, however, expected to be numbered differently, and to be rendered as separate sequences from clause sections.

8.6. Category label

Both section types can have a category label, which allows them to be semantically classified into recurring categories, which may be processed and rendered differently. Indicative values include:

- executive-summary,
- errata,
- methods,
- requirements,
- tests,
- reagents, and
- tests.

8.7. Restrictions on hanging elements

Generally, in standardization documents there is a strong need to reference text in clauses unambiguously by clause number, which makes hanging paragraphs undesirable.

Therefore, the StandardDocument model requires “hanging paragraphs” to be avoided. In particular, its sections should contain either blocks or subsections, but not both.

NOTE 1 The avoidance of “hanging paragraphs” is also a requirement in [ISO/IEC DIR 2:2018](#).

While this behavior is less pressing for Content sections (which are unnumbered), and Annexes (as exemplified in [ISO/IEC DIR 2:2018](#)), it is nonetheless not recommended.

8.8. Overview

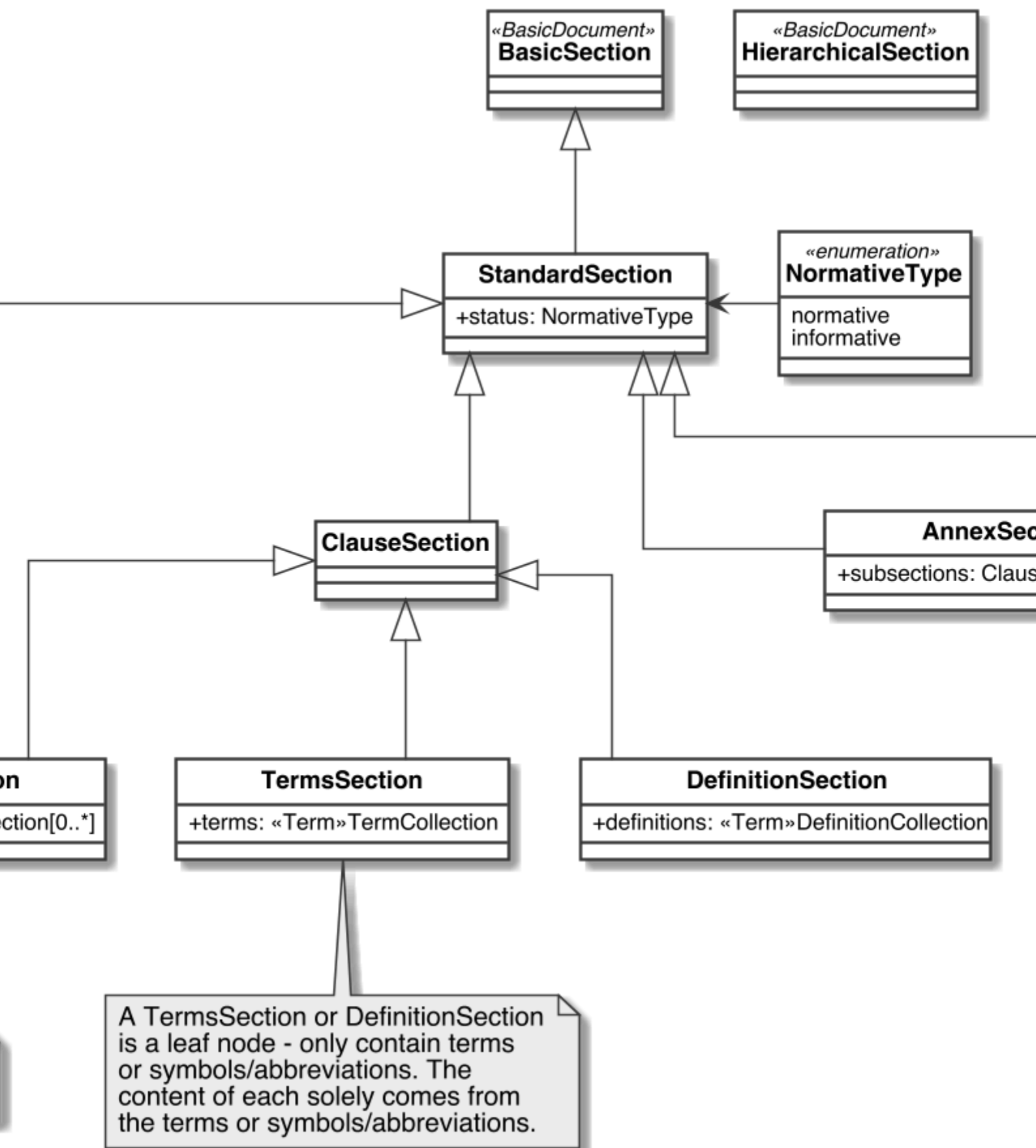


Figure 4 — StandardDocument model: Section

9. Citation and reference elements

9.1. Purpose

Standardization documents often cite other documents within, and as a result they usually contain a bibliography section indicating its data sources.

9.2. References section

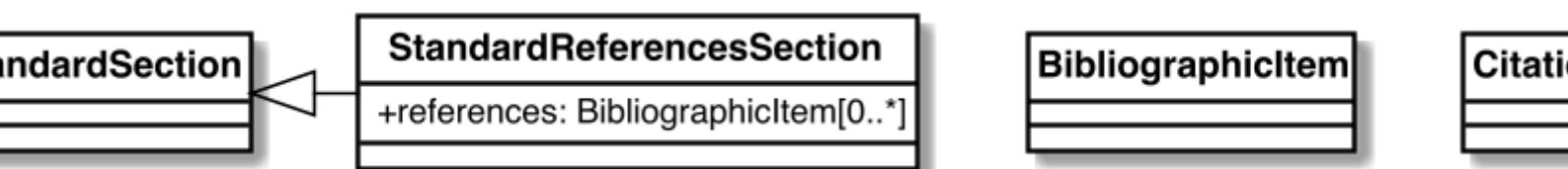


Figure 5 — Reference data models

9.3. StandardReferencesSection

The `StandardReferencesSection` class within *StandardDocument* is used to represent a bibliography section. It is used to collate references within the document, where there could be one or more of such sections within a document.

For example, some standardization documents differentiate normative or informative references, some split references into sections organized by concept relevance.

The `StandardReferencesSection` class is a specialized class of `StandardSection` ([Clause 8](#)).

Similar to the `ReferencesSection` of the *BasicDocument* model ([CC 36010](#)), they are leaf nodes which contain zero or more bibliographical references (as modelled in [CC 6900](#)), along with any prefatory text.

Table 6 — `StandardReferencesSection` attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
references	TODO: enum references's definition	O	N	BibliographicItem

9.4. BibliographicItem

`BibliographicItem` is defined in [CC 6900](#) as as a data model of [ISO 690 \(all parts\)](#) bibliographic items.

9.5. Citation

Citation is defined in [CC 6900](#) as as a data model of [ISO 690 \(all parts\)](#) citations.

9.6. Reference item

Each reference item is represented using the *BibliographicItem* class defined in [CC 6900](#).

It is not necessary to specialize reference items amongst documents from different standardization bodies as a document can cite documents of any type.

9.7. Citation

Each citation is represented by the *Citation* class defined in [CC 6900](#).

It is not necessary to specialize citations since it is simply a document data element reference linked to a reference item.

10. Terms and definition elements

10.1. Purpose

Standardization documents are used to convey normative information; to achieve so it is imperative to provide a standardized set of terms and definitions.

10.2. General

The *StandardDocument* model provides two further subclasses of clause sections (*ClauseSection*):

- Definition section, and
- Term section.

Both types of section can include prefatory text.

10.3. Definition sections

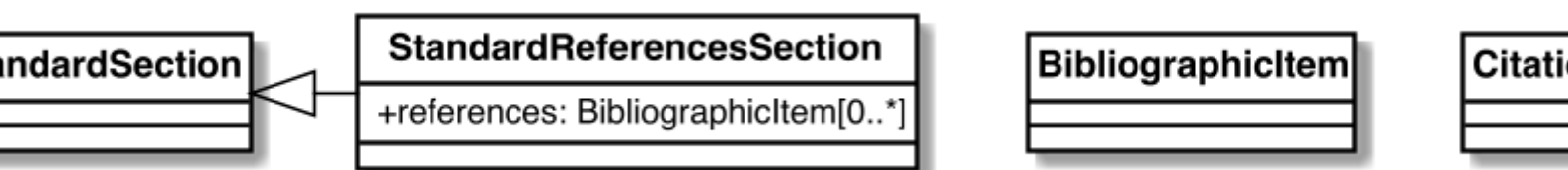


Figure 6 — Reference data models

10.4. StandardReferencesSection

The *StandardReferencesSection* class within *StandardDocument* is used to represent a bibliography section. It is used to collate references within the document, where there could be one or more of such sections within a document.

For example, some standardization documents differentiate normative or informative references, some split references into sections organized by concept relevance.

The *StandardReferencesSection* class is a specialized class of *StandardSection* ([Clause 8](#)).

Similar to the *ReferencesSection* of the *BasicDocument* model ([CC 36010](#)), they are leaf nodes which contain zero or more bibliographical references (as modelled in [CC 6900](#)), along with any prefatory text.

Table 7 — *StandardReferencesSection* attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
references	TODO: enum references's definition	O	N	BibliographicItem

10.5. BibliographicItem

BibliographicItem is defined in [CC 6900](#) as as a data model of [ISO 690 \(all parts\)](#) bibliographic items.

10.6. Citation

Citation is defined in [CC 6900](#) as as a data model of [ISO 690 \(all parts\)](#) citations.

10.7. Term sections

10.8. Term

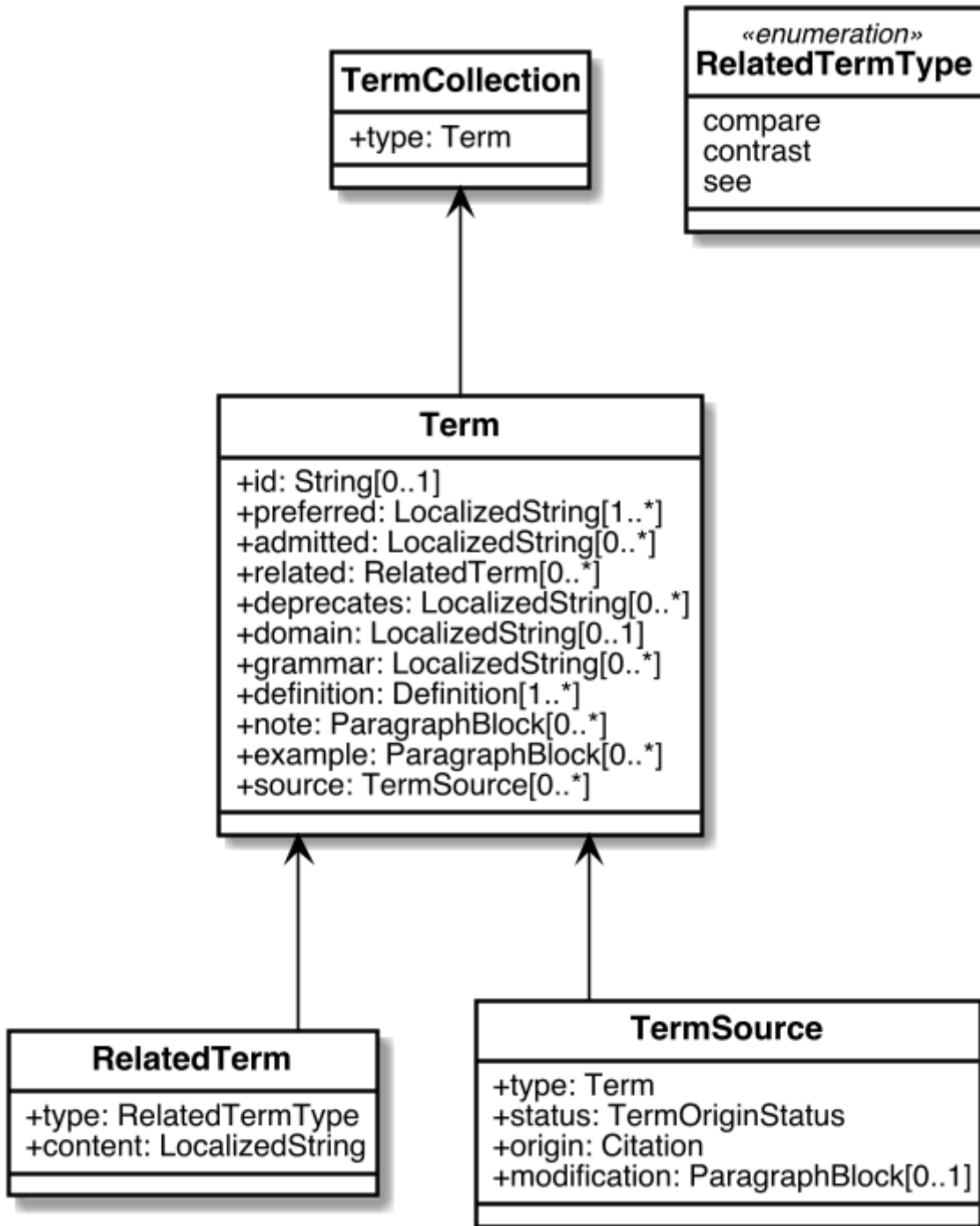


Figure 7 — Term class data models

10.9. TermCollection

Table 8 — TermCollection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	TODO: enum type's definition	M	1	Term

10.10. Term

A term (Term) represents a terminology entry with its definition.

NOTE 1 The Term definition fully aligns with the structure and requirements of terms in [ISO/IEC DIR 2:2018](#).

Table 9 – Term attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
id	An optional identifier for the term, to be used in cross-references.	O	1	String
preferred	One or more names under which the term being defined is canonically known.	M	N	LocalizedString
admitted	Zero or more names which are acceptable synonyms for the term being defined.	O	N	LocalizedString
related	Zero or more names which are related to the term being defined. Each has a type, indicating how the term is related; permitted values are compare (for “see also” references to terms), contrast (for terms that illuminate the term definition as what it is not), see (if this is a deprecated term, to reference the preferred term)	O	N	RelatedTerm
deprecates	Zero or more names which are deprecated synonyms for the term being defined.	O	N	LocalizedString
domain	An optional semantic domain for the term being defined, in case the term is ambiguous between several semantic domains.	O	1	LocalizedString
grammar	Zero or more abbreviations giving grammatical information about the term (e.g. part of speech, gender)	O	N	LocalizedString
definition	The definition of the term applied in the current document.	M	N	Definition
note	Zero or more notes about the term.	O	N	ParagraphBlock
example	Zero or more examples of how the term is to be used.	O	N	ParagraphBlock
source	Zero or more bibliographical sources for the term. These include the origin of the term, which is its bibliographical citation (as defined in CC 6900); the status of the definition (whether identical to	O	N	TermSource

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
	the definition given in the origin cited, or modified); and, if the definition is modified, a description of the modification to the definition applied for this document.			

10.11. RelatedTerm

Table 10 — RelatedTerm attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	TODO: enum type's definition	M	1	RelatedTermType
content	TODO: enum content's definition	M	1	LocalizedString

10.12. RelatedTermType

Table 11 — RelatedTermType values

Name	Definition
compare	
contrast	
see	

10.13. TermSource

Table 12 — TermSource attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	TODO: enum type's definition	M	1	Term
status	TODO: enum status's definition	M	1	TermOriginStatus
origin	TODO: enum origin's definition	M	1	Citation
modification	TODO: enum modification's definition	O	1	ParagraphBlock

10.14. Nested sections

Both definition and term sections are leaf nodes, and cannot contain subclauses of their own.

Nested definition and term sections should be modelled as ClauseSections with subclauses, terminating in definition and term sections.

10.15. Overview

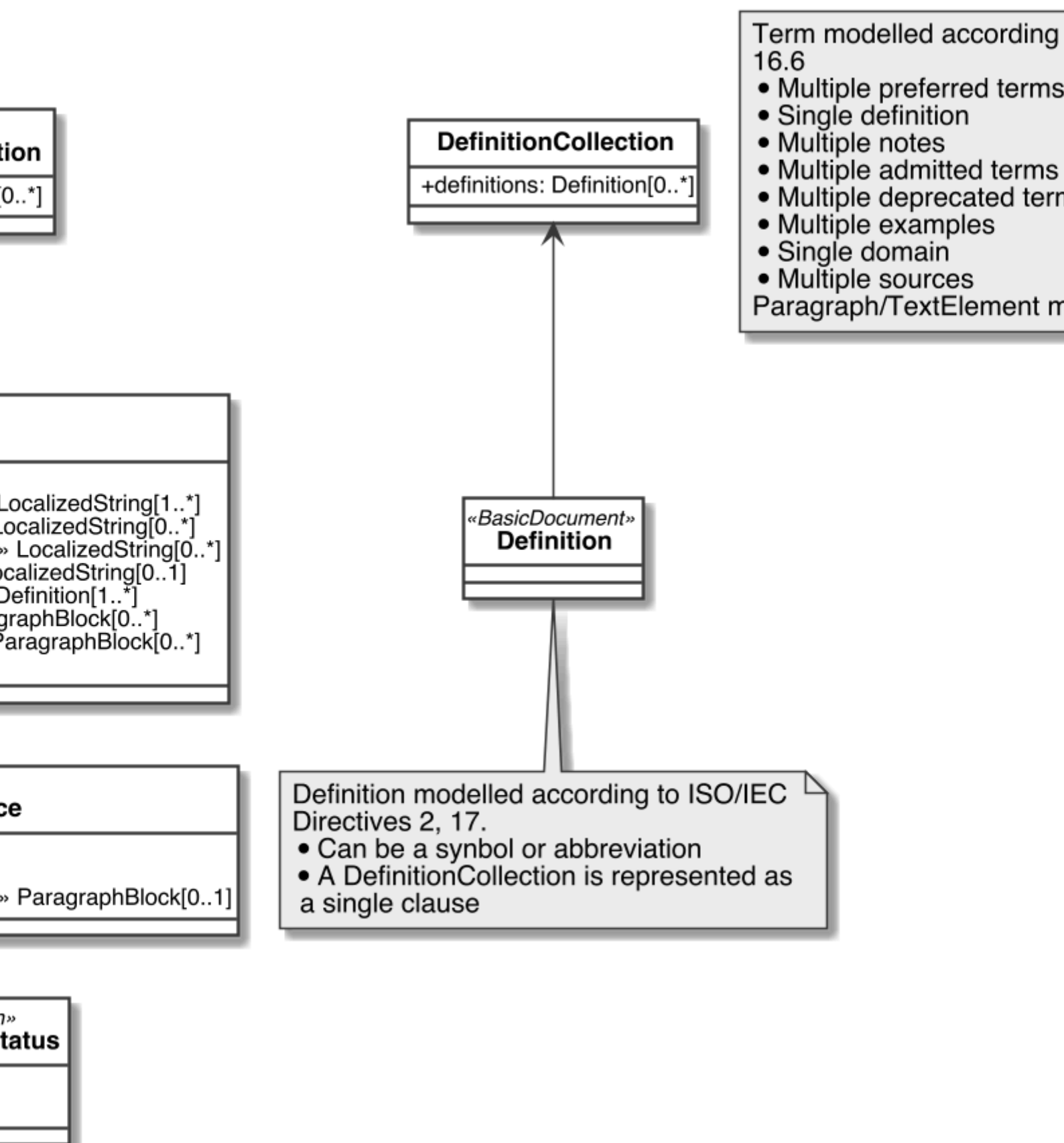


Figure 8 — StandardDocument model: Term Section and Definition Section

11. Inline elements

The inline elements of StandardDocument adhere to the inline element models from BasicDocument.

Specializations and profiles of the StandardDocument model may specialize the modelling of inline elements.

12. Block

The block elements of StandardDocument adhere to the block element BasicDocument model and the StandardDocument model.

Specializations and profiles of the StandardDocument model may specialize the modelling of blocks.

13. Data types

Data types in StandardDocument models are fully inherited from those of the BasicDocument model.

Specializations and profiles of the StandardDocument model may specialize and extend the modelling of data types.

Annex A.. Profile: ISO standardization deliverables

A.1.. General

This profile provides the specialization of the *StandardDocument* document metamodel for ISO standardization deliverables described in [ISO/IEC DIR 2:2018](#).

The *IsoStandardDocument* model fully covers all ISO standardization deliverables, including International Standards, Guides, Technical Reports, Technical Specifications, International Workshop Agreements, Amendments and Corrigenda.

A.2.. Document and section specialization

The *IsoStandardDocument* model extends the *StandardDocument* modelling of the document by requiring the following specific types of section:

- A *Preface*, consisting of a mandatory *Foreword* (a *BasicSection*: [CC 36010](#)), and an optional *Introduction* (a *ClauseSection*: [Clause 8](#), since an ISO Introduction can optionally have numbered subsections).
- One or more *Clause Sections* ([Clause 8](#)).
- Zero or more *Annexes* ([Clause 8](#)).

Annexes in the *IsoStandardDocument* can optionally contain one or more *Appendixes*, which can be thought of as annexes of the annex section.

NOTE 1 Appendixes are not defined in [ISO/IEC DIR 2:2018](#), but they appear in the companion standard [ISO/IEC DIR 1](#). For example, Annex SL of [ISO/IEC DIR 1](#) includes Appendix 1, Appendix 2, and Appendix 3; Appendix 2 “High level structure, identical core text, common terms and core definitions” itself contains an Introduction and 10 clauses, including a Terms clause.

The *IsoStandardDocument* model document also includes zero or more *termSources* elements, giving the citations for the documents that the Terms Section draws on for its definitions and repertoire of terms.

The *IsoStandardDocument* model document uses ISO-specific extensions to the [CC 6900](#) metadata about a document ([A.9](#)).

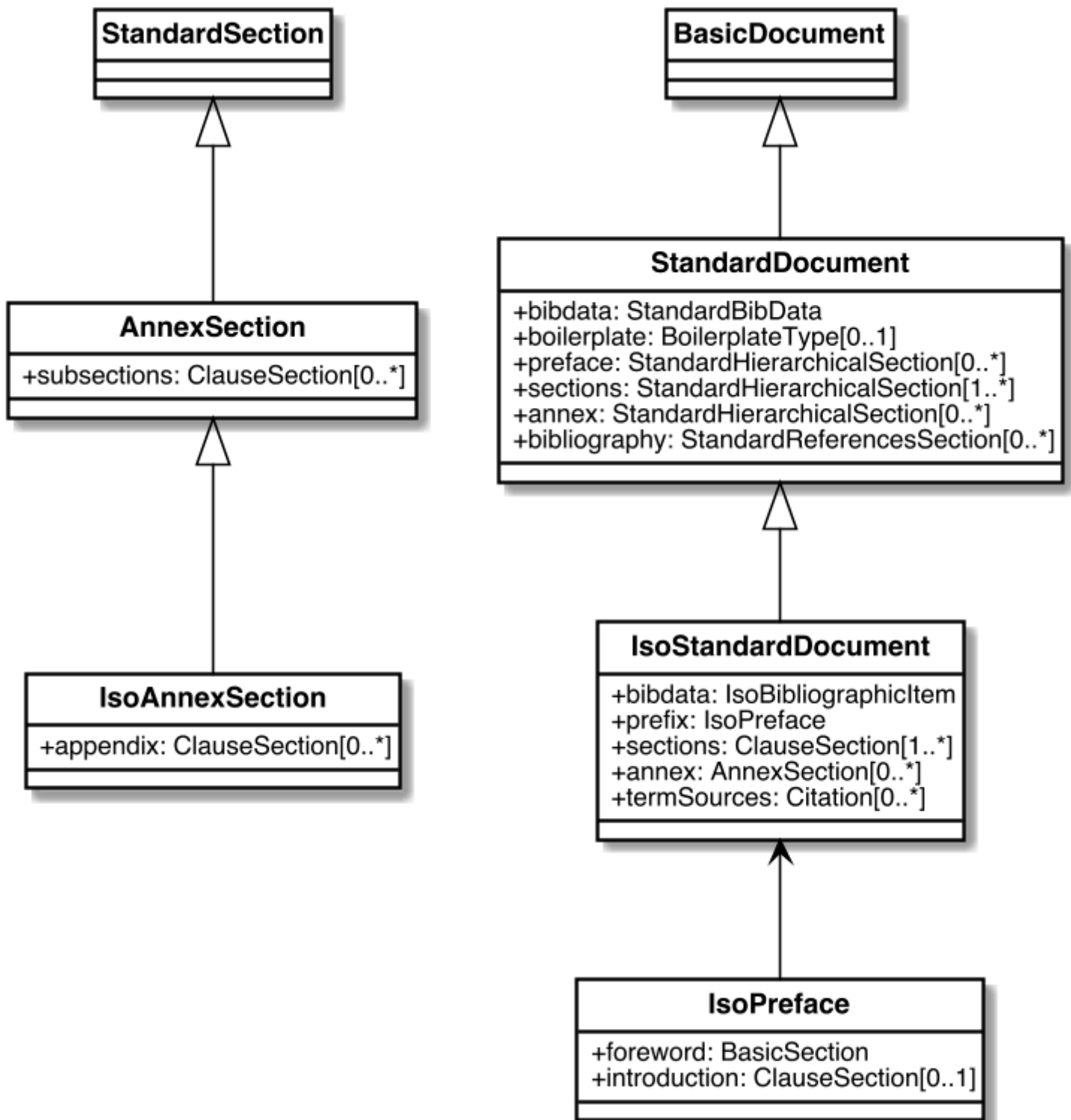


Figure A.1 — ISO Standard Document

A.3.. IsoStandardDocument

Table A-1 — IsoStandardDocument attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
bibdata	TODO: enum bibdata's definition	M	1	IsoBibliographicItem

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
prefix	TODO: enum prefix's definition	M	1	IsoPreface
sections	TODO: enum sections's definition	M	N	ClauseSection
annex	TODO: enum annex's definition	O	N	AnnexSection
termSources	TODO: enum termSources's definition	O	N	Citation

A.4.. IsoAnnexSection

Table A-1 — IsoAnnexSection attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
appendix	TODO: enum appendix's definition	O	N	ClauseSection

A.5.. IsoPreface

Table A-1 — IsoPreface attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
foreword	TODO: enum foreword's definition	M	1	BasicSection
introduction	TODO: enum introduction's definition	O	1	ClauseSection

A.6.. Block specialization

The IsoStandardDocument model extends the StandardDocument modelling of blocks for one block definition.

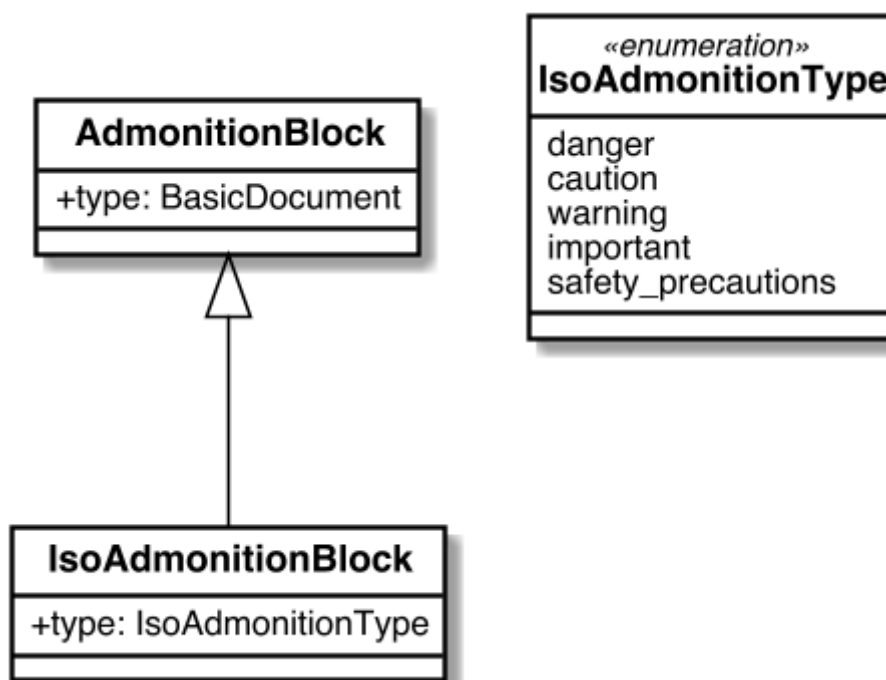


Figure A.1 — ISO Standard Document blocks

A.7.. IsoAdmonitionBlock

The types that an Admonition can belong to are overridden to belong to the following set prescribed in [ISO/IEC DIR 2:2018](#):

- Danger,
- Caution,
- Warning,
- Important,
- Safety Precaution.

Table A-1 — IsoAdmonitionBlock attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	The type of admonition.	M	1	IsoAdmonitionType

A.8.. IsoAdmonitionType

Table A-1 — IsoAdmonitionType values

Name	Definition
danger	Danger
caution	Caution
warning	Warning
important	Important
safety_precautions	Safety precaution

A.9.. Document metadata specialization

The IsoBibliographicItem model customizes parts of BibliographicItem to fully cover representations of ISO standardization deliverables' metadata.

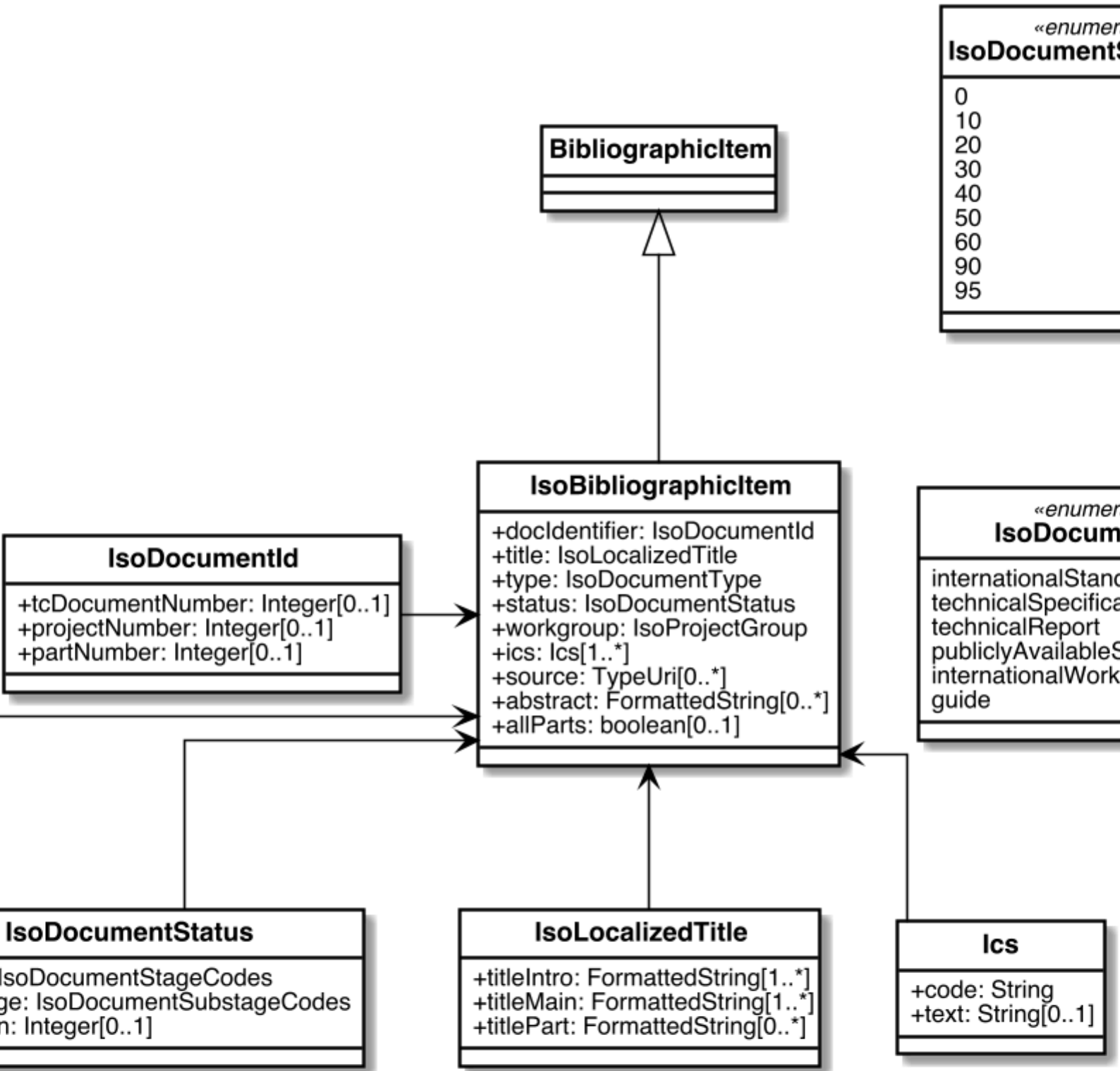
While `BibliographicItem` is intended as a highly generic model, however, it is useful for the metadata describing the current document to be enriched to provide metadata relevant to the standard class.

`IsoBibliographicItem` is enriched compared to `BibliographicItem` as follows:

- The document identifier is refined to contain three optional parts:
 - a project number,
 - a part number, and
 - a technical committee document number.
- The document title is broken down into:
 - an optional Introductory component,
 - a mandatory Main component, and
 - an optional Part component.
- The document type is constrained to be one of the following:
 - International Standard,
 - Technical Specification,
 - Technical Report,
 - Publicly Available Specification,
 - International Workshop Agreement, and
 - Guide.
- The document status is broken down into:
 - an ISO Document Stage code ([ISO International harmonized stage codes](#)),
 - an ISO Document Substage code ([ISO International harmonized stage codes](#)), and
 - an Iteration number.
- The editorial group responsible for the document is defined with four components:
 - a mandatory Technical Committee,
 - an optional Subcommittee,
 - an optional Workgroup, and
 - an optional Secretariat.

The Technical Committee, Subcommittee and Workgroup, in turn, are instances of the ISO Subgroup class, defined as having a name, an optional type, and an optional number.

- The document has one or more International Classification for Standards classifications [ISO International Classification for Standards](#). The ICS classification is defined as a code, and an optional text explaining the code.
- The document has zero or more source URIs.
- The document has an optional abstract.
- The document has an optional “All Parts” boolean attribute, indicating whether the bibliographic description applies across all Part documents under the same project number (which makes it a multi-document description).



A.10.. IsoBibliographicItem**Table A-1 — IsoBibliographicItem attributes**

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
docIdentifier	TODO: enum docIdentifier's definition	M	1	IsoDocumentId
title	TODO: enum title's definition	M	1	IsoLocalizedTitle
type	TODO: enum type's definition	M	1	IsoDocumentType
status	TODO: enum status's definition	M	1	IsoDocumentStatus
workgroup	TODO: enum workgroup's definition	M	1	IsoProjectGroup
ics	TODO: enum ics's definition	M	N	Ics
source	TODO: enum source's definition	O	N	TypeId
abstract	TODO: enum abstract's definition	O	N	FormattedString
allParts	TODO: enum allParts's definition	O	1	boolean

A.11.. IsoDocumentId**Table A-1 — IsoDocumentId attributes**

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
tcDocumentNumber	TODO: enum tcDocumentNumber's definition	O	1	Integer
projectNumber	TODO: enum projectNumber's definition	O	1	Integer
partNumber	TODO: enum partNumber's definition	O	1	Integer

A.12.. IsoDocumentType**Table A-1 — IsoDocumentType values**

Name	Definition
internationalStandard	
technicalSpecification	
technicalReport	
publiclyAvailableSpecification	
internationalWorkshopAgreement	
guide	

A.13.. IsoDocumentStatus**Table A-1 — IsoDocumentStatus attributes**

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
stage	TODO: enum stage's definition	M	1	IsoDocumentStageCodes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
substage	TODO: enum substage's definition	M	1	IsoDocumentSubstageCodes
iteration	TODO: enum iteration's definition	O	1	Integer

A.14.. IsoDocumentStageCodes

Table A-1 — IsoDocumentStageCodes values

Name	Definition
0	
10	
20	
30	
40	
50	
60	
90	
95	

A.15.. IsoDocumentSubstageCodes

Table A-1 — IsoDocumentSubstageCodes values

Name	Definition
0	
20	
60	
90	
92	
93	
98	
99	

A.16.. IsoLocalizedTitle

Table A-1 — IsoLocalizedTitle attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
titleIntro	TODO: enum titleIntro's definition	M	N	FormattedString
titleMain	TODO: enum titleMain's definition	M	N	FormattedString
titlePart	TODO: enum titlePart's definition	O	N	FormattedString

A.17.. IsoProjectGroup

Table A-1 — IsoProjectGroup attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
technicalCommittee	TODO: enum technicalCommittee's definition	M	1	IsoSubGroup
subcommittee	TODO: enum subcommittee's definition	O	1	IsoSubGroup

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
workgroup	TODO: enum workgroup's definition	O	1	IsoSubGroup
secretariat	TODO: enum secretariat's definition	O	1	String

A.18.. IsoSubGroup**Table A-1 — IsoSubGroup attributes**

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	TODO: enum type's definition	O	1	String
number	TODO: enum number's definition	O	1	Integer
name	TODO: enum name's definition	M	1	String

Annex B.. Profile: CalConnect standardization deliverables

B.1.. General

This profile provides the specialization of the *StandardDocument* document metamodel for CalConnect standardization deliverables described in [CC/DIR 10002:2019](#).

The CsdStandardDocument model fully covers all CalConnect standardization deliverables, including Standards, Reports, Specifications, Directives, Guides, Amendments and Corrigenda.

B.2.. Document and section specialization

The CsdStandardDocument model extends the StandardDocument modelling of the document by requiring the following specific types of section:

- An optional *Foreword* (a StandardsHierarchicalSection, see [Clause 8](#)).
- An optional *Introduction* (a StandardsHierarchicalSection, see [Clause 8](#)).
- One or more clause sections (see [Clause 8](#)).
- Zero or more Annexes (see [Clause 8](#)).

B.3.. Block specialization

The Block model is specialized:

- The definition of Figure is altered to allow ASCII artwork as an option for its content, as a string (pre). As with the sourcecode element ([CC 36010](#)), the string should be treated as pre-formatted text, with whitespace treated as significant.

B.4.. Inline elements specialization

The CsdStandardDocument model extends the StandardDocument modelling of inline elements by one element:

- The keyword element is added as a subclass of Text Element ([CC 36010](#)). The element is used to represent code keywords, as with code in HTML, and should be rendered in monospace.

B.5.. Document metadata specialization

The bibliographic modelling of CalConnect documents is further enriched compared to BibliographicItem as follows:

- The document type is constrained to be one of the following:
 - Standards,
 - Reports,
 - Specifications,
 - Directives,
 - Guides,
 - Amendments,
 - Corrigenda and Administrative.
- The document status is constrained to be one of the following:
 - Proposal,
 - Working Draft,
 - Committee Draft,
 - Final Draft,
 - Published,
 - Withdrawn.

- The editorial group responsible for the document is defined with a single components: a mandatory Technical Committee. The Technical Committee has a type of either Technical or Provisional.

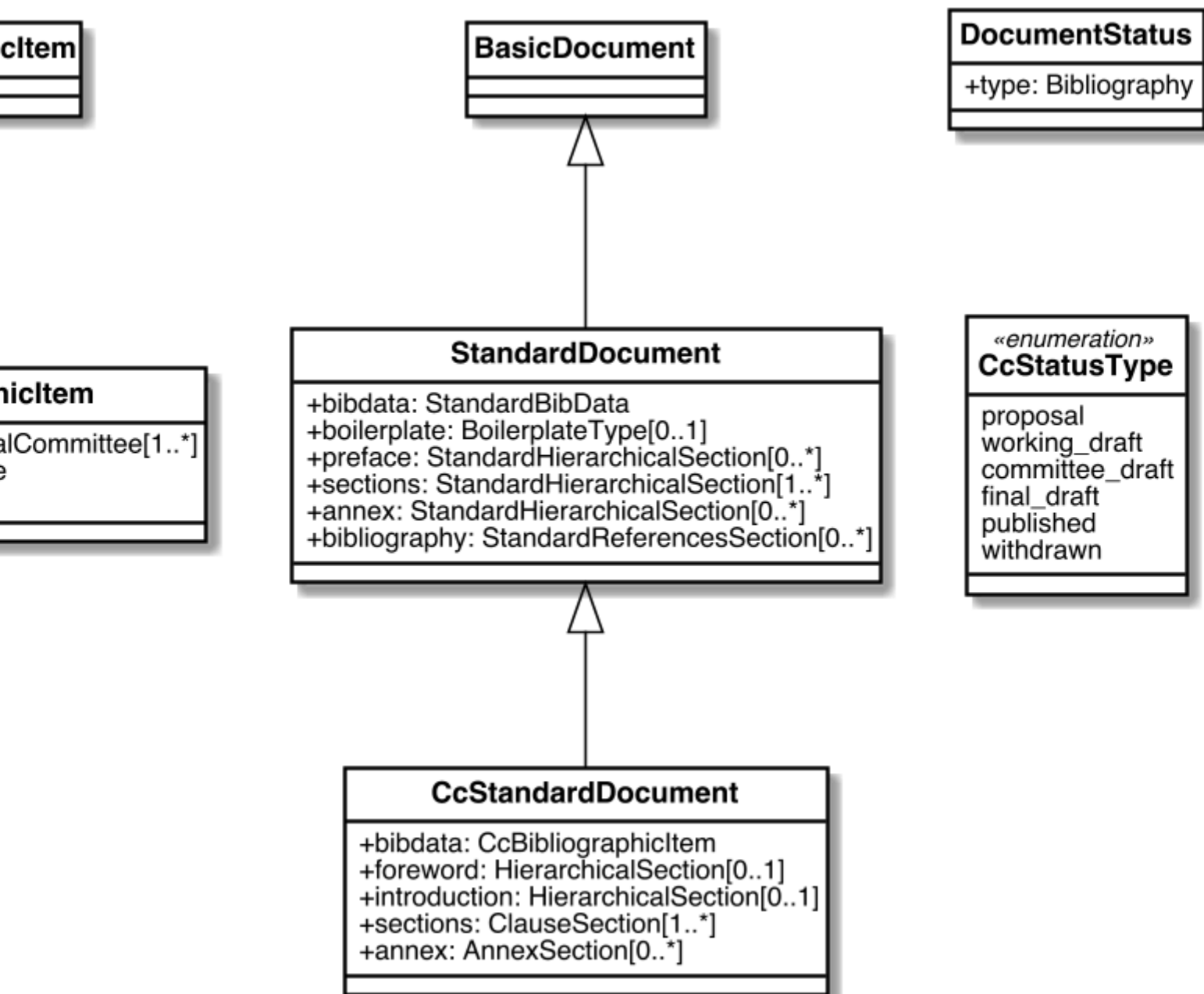


Figure B.1 — CalConnect Standard document class data model

B.6.. CcBibliographicItem

Table B-1 — CcBibliographicItem attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
committee	TODO: enum committee's definition	M	N	CcTechnicalCommittee
type	TODO: enum type's definition	M	1	CcDocumentType

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
title	TODO: enum title's definition	M	1	FormattedString

B.7.. CcDocumentType

Table B-1 — CcDocumentType values

Name	Definition
directive	
standard	
guide	
specification	
amendment	
technical_corrigendum	
report	
administrative	

B.8.. CcStandardDocument

Table B-1 — CcStandardDocument attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
bibdata	TODO: enum bibdata's definition	M	1	CcBibliographicItem
foreword	TODO: enum foreword's definition	O	1	HierarchicalSection
introduction	TODO: enum introduction's definition	O	1	HierarchicalSection
sections	TODO: enum sections's definition	M	N	ClauseSection
annex	TODO: enum annex's definition	O	N	AnnexSection

B.9.. CcStatusType

Table B-1 — CcStatusType values

Name	Definition
proposal	
working_draft	
committee_draft	
final_draft	
published	
withdrawn	

B.10.. CcTechnicalCommittee

Table B-1 — CcTechnicalCommittee attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	TODO: enum type's definition	M	1	CcTechnicalCommitteeType
name	TODO: enum name's definition	M	1	String

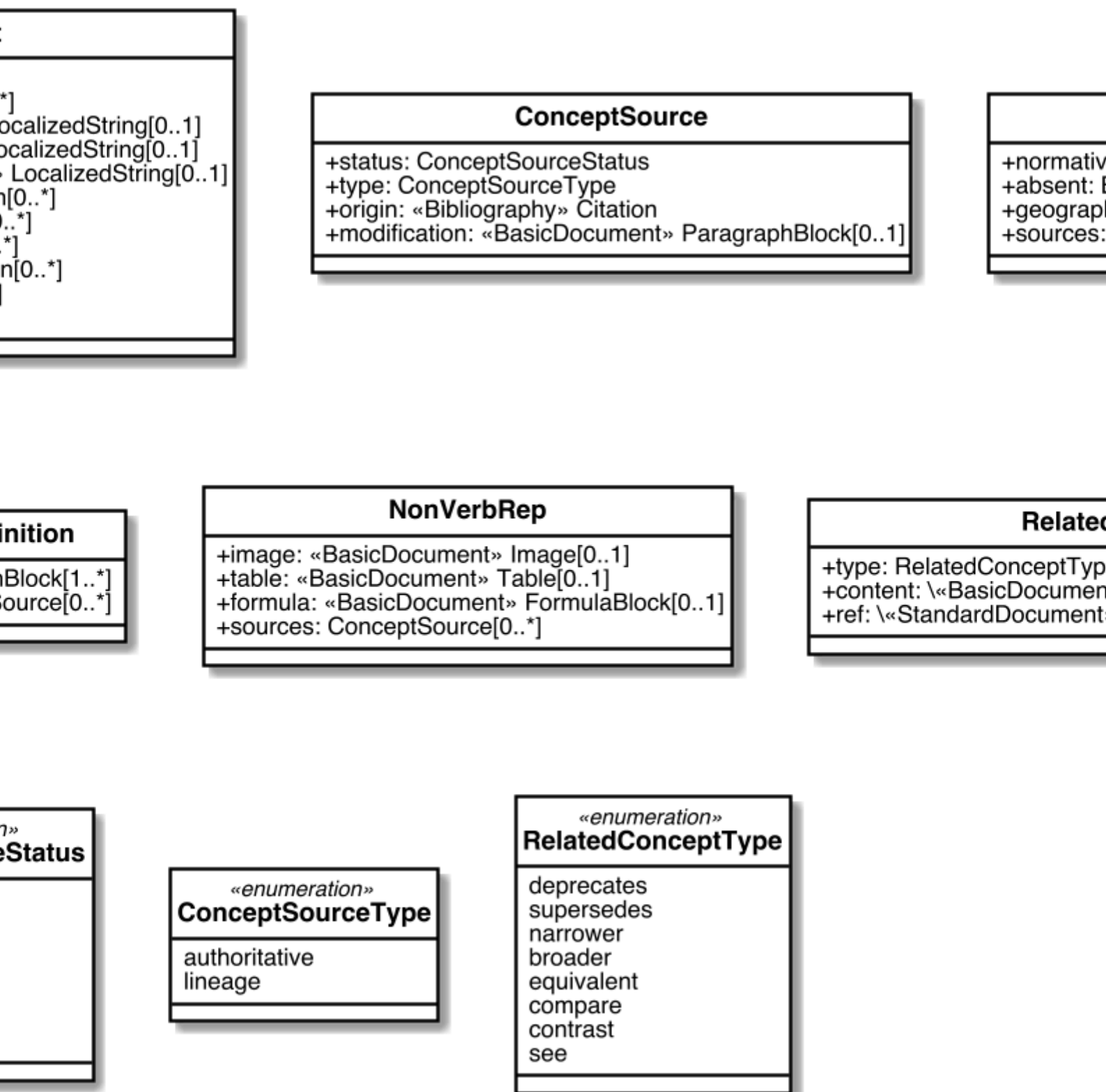
B.11.. CcTechnicalCommitteeType**Table B-1 — CcTechnicalCommitteeType values**

Name	Definition
technical	
provisional	

Annex C.. Concept and concept collection models

C.1.. General

This annex provides an informative reference model for management of concepts and concept collections, compliant with ISO 10241-1 and ISO 704.

C.2.. Concept**Figure C.1 — Concept models**

C.3.. Concept

Table C-1 — Concept attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
id	TODO: enum id's definition	O	1	String
designations	TODO: enum designations's definition	M	N	Designation
domain	TODO: enum domain's definition	O	1	Clause 5.1 LocalizedString
subject	TODO: enum subject's definition	O	1	Clause 5.1 LocalizedString
usagelInfo	TODO: enum usagelInfo's definition	O	1	Clause 5.1 LocalizedString
definition	TODO: enum definition's definition	O	N	GlossaristDefinition
nonverbalrep	TODO: enum nonverbalrep's definition	O	N	NonVerbRep
notes	TODO: enum notes's definition	O	N	GlossaristDefinition
examples	TODO: enum examples's definition	O	N	GlossaristDefinition
sources	TODO: enum sources's definition	O	N	ConceptSource
related	TODO: enum related's definition	O	N	RelatedConcept

C.4.. GlossaristDefinition

Table C-1 — GlossaristDefinition attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
content	TODO: enum content's definition	M	N	ParagraphBlock
sources	TODO: enum sources's definition	O	N	ConceptSource

C.5.. NonVerbRep

Table C-1 — NonVerbRep attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
image	TODO: enum image's definition	O	1	Clause 5.1 Image
table	TODO: enum table's definition	O	1	Clause 5.1 Table
formula	TODO: enum formula's definition	O	1	Clause 5.1 FormulaBlock
sources	TODO: enum sources's definition	O	N	ConceptSource

C.6.. RelatedConcept

Table C-1 — RelatedConcept attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
type	TODO: enum type's definition	M	1	RelatedConceptType
content	TODO: enum content's definition	M	1	<<BasicDocument>> LocalizedString
ref	TODO: enum ref's definition	O	1	<<StandardDocument>> ReferenceToTermbase

C.7.. RelatedConceptType

Table C-1 — RelatedConceptType values

Name	Definition
deprecates	
supersedes	
narrower	
broader	
equivalent	
compare	
contrast	
see	

C.8.. Concept sources

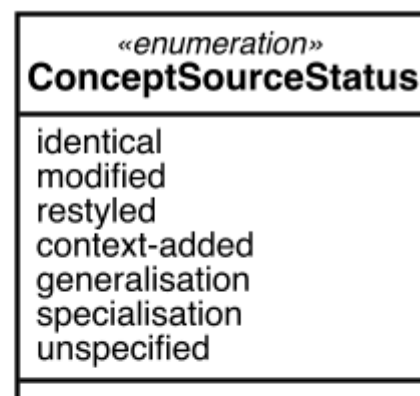
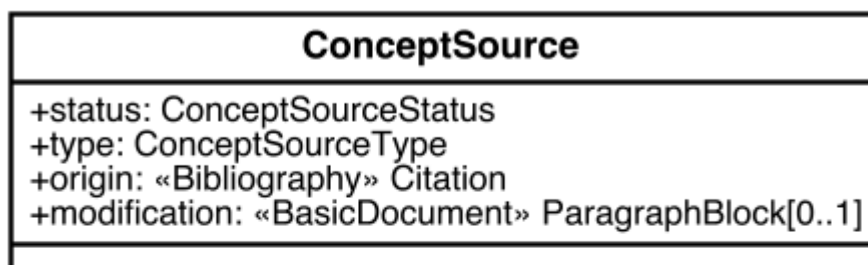


Figure C.1 — Concept source related classes

C.9.. ConceptSource

Table C-1 — ConceptSource attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
status	TODO: enum status's definition	M	1	ConceptSourceStatus
type	TODO: enum type's definition	M	1	ConceptSourceType
origin	TODO: enum origin's definition	M	1	Citation
modification	TODO: enum modification's definition	O	1	Clause 5.1 ParagraphBlock

C.10.. ConceptSourceStatus

Table C-1 — ConceptSourceStatus values

Name	Definition
identical	
modified	
restyled	
context-added	
generalisation	
specialisation	
unspecified	

C.11.. ConceptSourceType

Table C-1 — ConceptSourceType values

Name	Definition
authoritative	
lineage	

C.12.. Designation

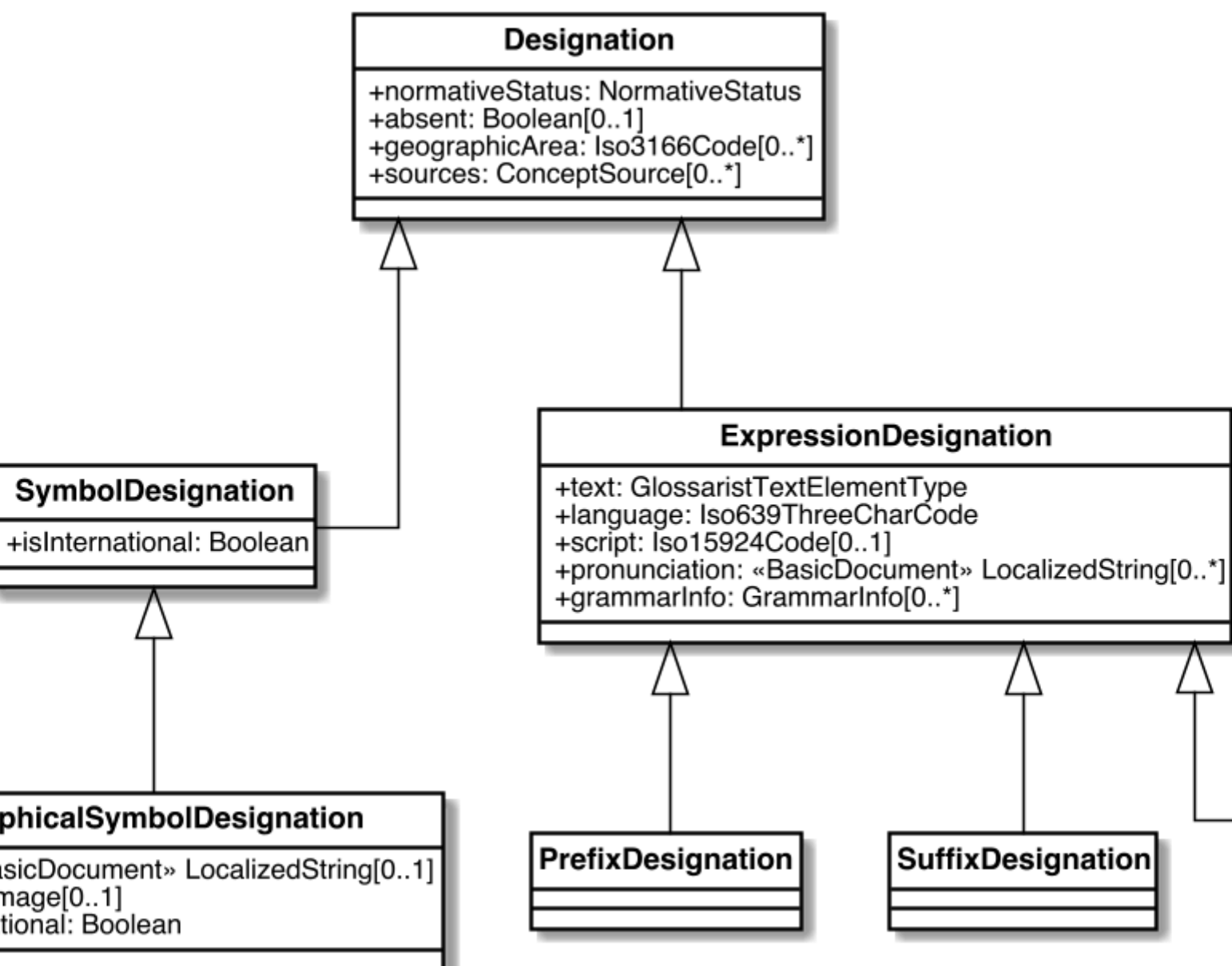


Figure C.1 — Designation models

C.13.. Designation

Table C-1 — Designation attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
normativeStatus	TODO: enum normativeStatus's definition	M	1	NormativeStatus
absent	TODO: enum absent's definition	O	1	Boolean
geographicArea	TODO: enum geographicArea's definition	O	N	Iso3166Code

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
sources	TODO: enum sources's definition	O	N	ConceptSource

C.14.. NormativeStatus

Table C-1 — NormativeStatus values

Name	Definition
preferred	
admitted	
deprecated	

C.15.. ExpressionDesignation

Table C-1 — ExpressionDesignation attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
text	TODO: enum text's definition	M	1	GlossaristTextElementType
language	TODO: enum language's definition	M	1	Iso639ThreeCharCode
script	TODO: enum script's definition	O	1	Iso15924Code
pronunciation	TODO: enum pronunciation's definition	O	N	Clause 5.1 LocalizedString
grammarInfo	TODO: enum grammarInfo's definition	O	N	GrammarInfo

C.16.. GrammarInfo

Table C-1 — GrammarInfo attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
gender	TODO: enum gender's definition	O	N	GrammarGender
isPreposition	TODO: enum isPreposition's definition	O	1	Boolean
isParticiple	TODO: enum isParticiple's definition	O	1	Boolean
isAdjective	TODO: enum isAdjective's definition	O	1	Boolean
isVerb	TODO: enum isVerb's definition	O	1	Boolean
isAdverb	TODO: enum isAdverb's definition	O	1	Boolean
isNoun	TODO: enum isNoun's definition	O	1	Boolean

C.17.. SymbolDesignation**Table C-1 — SymbolDesignation attributes**

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
isInternational	TODO: enum isInternational's definition	M	1	Boolean

C.18.. PrefixDesignation**C.19.. SuffixDesignation****C.20.. AbbreviationDesignation****Table C-1 — AbbreviationDesignation attributes**

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
isInternational	TODO: enum isInternational's definition	M	1	Boolean
abbreviationType	TODO: enum abbreviationType's definition	O	1	AbbreviationType

C.21.. AbbreviationType**Table C-1 — AbbreviationType values**

Name	Definition
truncation	
acronym	
initialism	

C.22.. LetterSymbolDesignation**Table C-1 — LetterSymbolDesignation attributes**

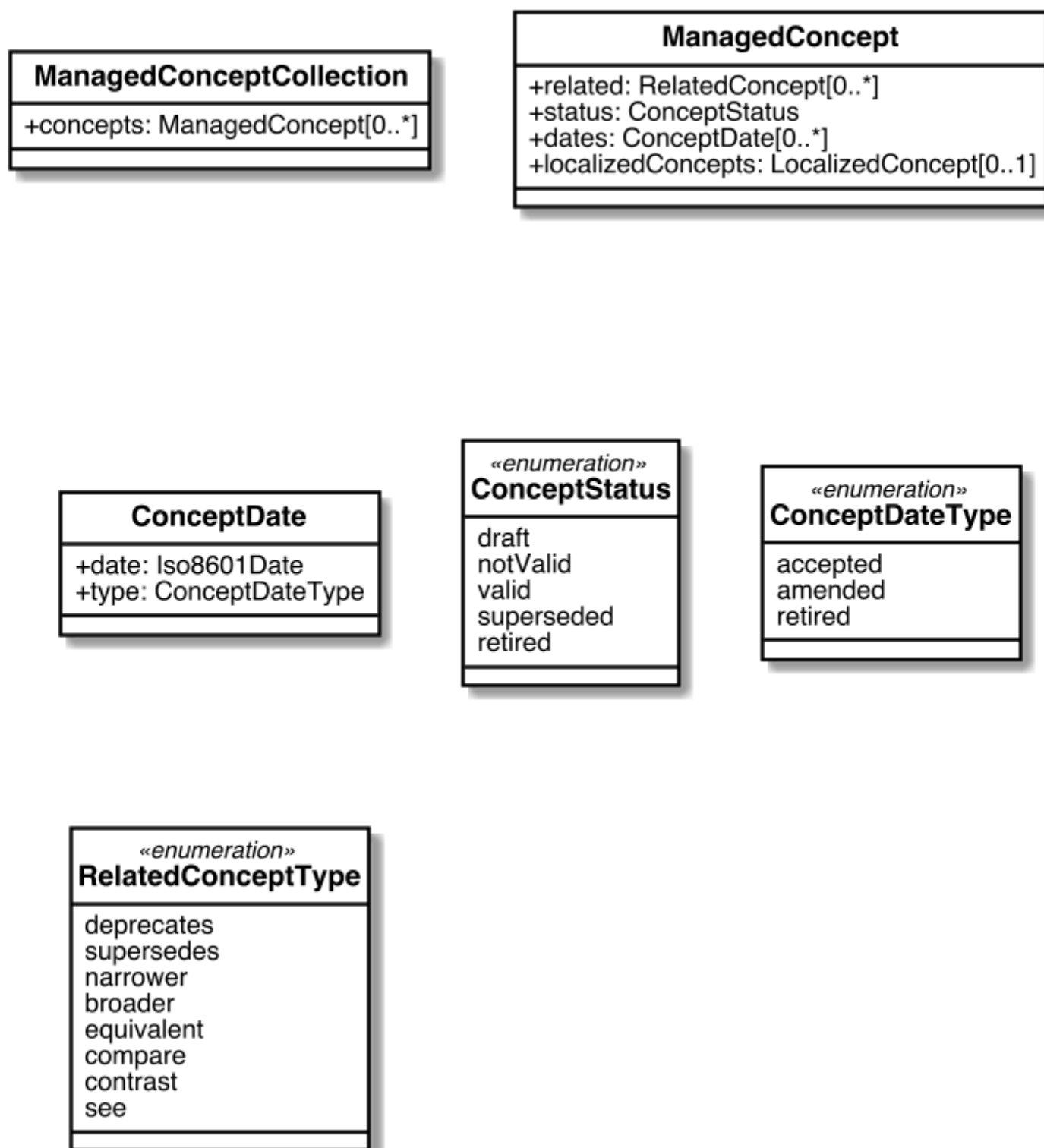
Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
text	TODO: enum text's definition	M	1	Clause 5.1 LocalizedString
language	TODO: enum language's definition	M	1	Iso639ThreeCharCode
script	TODO: enum script's definition	O	1	Iso15924Code

C.23.. GraphicalSymbolDesignation**Table C-1 — GraphicalSymbolDesignation attributes**

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
text	TODO: enum text's definition	O	1	Clause 5.1 LocalizedString
image	TODO: enum image's definition	O	1	Image
isInternational	TODO: enum isInternational's definition	M	1	Boolean

C.24.. GrammarGender**Table C-1 — GrammarGender values**

Name	Definition
masculine	
feminine	
neuter	
common	

C.25.. Managed concepts and collection**Figure C.1 — Managed concepts and collection classes**

C.26.. ManagedConcept

Table C-1 — ManagedConcept attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
related	TODO: enum related's definition	O	N	RelatedConcept
status	TODO: enum status's definition	M	1	ConceptStatus
dates	TODO: enum dates's definition	O	N	ConceptDate
localizedConcepts	TODO: enum localizedConcepts's definition	O	1	LocalizedConcept

C.27.. ConceptStatus

Table C-1 — ConceptStatus values

Name	Definition
draft	
notValid	
valid	
superseded	
retired	

C.28.. LocalizedConcept

Table C-1 — LocalizedConcept attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
language	TODO: enum language's definition	M	1	Iso639ThreeCharCode
script	TODO: enum script's definition	O	1	Iso15924Code
concept	TODO: enum concept's definition	M	1	Concept

C.29.. ConceptDate

Table C-1 — ConceptDate attributes

Name	Definition	Mandatory/ Optional/ Conditional	Max Occur	Data Type
date	TODO: enum date's definition	M	1	Iso8601Date
type	TODO: enum type's definition	M	1	ConceptDateType

C.30.. ConceptDateType

Table C-1 — ConceptDateType values

Name	Definition
accepted	
amended	
retired	

C.31.. RelatedConceptType

Table C-1 — RelatedConceptType values

Name	Definition
deprecates	

Name	Definition
supersedes	
narrower	
broader	
equivalent	
compare	
contrast	
see	

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